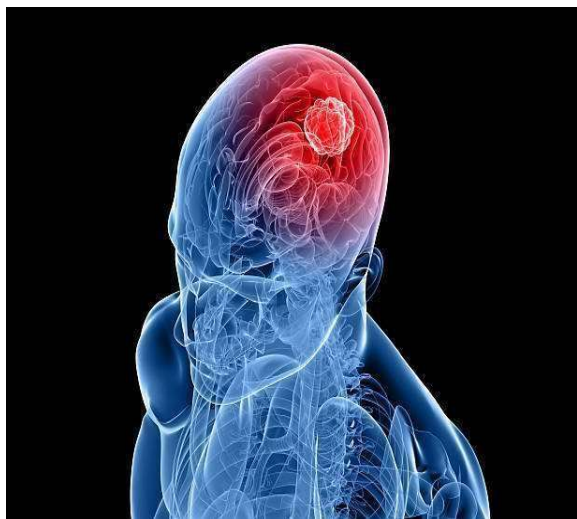




BRAIN TUMOR **DETECTION USING** **IMAGE SEGMENTATION**

DSP PROJECT

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PROJECT FINAL REPORT

- **PROBLEM?**

Brain tumor detection and classification using MRI images is the most troublesome task within the space of medicative field. Magnetic resonance imaging may be a procedure, typically adopted by the medical specialist for illustration of inner structure of the build with no surgery. It provides long information concerning the human delicate tissue that helps within the conclusion of brain tumor.

The major challenge faced was to locate and extract the proper tumor region from the image. Due to several lighting issues, unnecessary white portions were present in the image which could wrongly be segmented as a tumor. Also the unwanted noise and reduced contrast displays several regions from the image that are falsely claimed as a tumor. So in this project, we are using a technique that confirms the presence of tumor and then it focus on evaluating its stage, i.e., benign or malignant, by coping up with all above mentioned challenges.

- **INTRODUCTION**

We have used two different methodologies to segment a tumor from an MRI image and determine its type, for this one segmentation and one clustering technique has been implemented. Applying these will allow us to determine the most appropriate method to segment the tumor from each of the images.

- **DSP TECHNIQUES USED**

1. De-noising method
2. Filtering
3. Image Analysis

- **PROPOSED METHODOLOGY**

In this project, we have described our objective in two parts, the first half deals with detection of brain tumor that is the presence of the tumor in the provided MRI. The other part contains the classification of the tumor.

- **BLOCK DIAGRAM**



- **WORKING EXPLANATION**

1. MRI OF BRAIN IMAGES:

In the first step, data is been provided in the form of magnetic resonance images (MRI) that are been collected in their original format **.dcm**. We have used the **dicomread** and **dicominfo** commands from MATLAB to read and display the information of MRI images respectively.

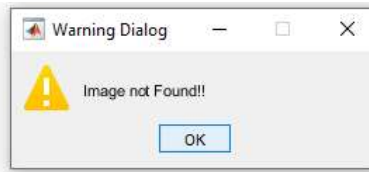
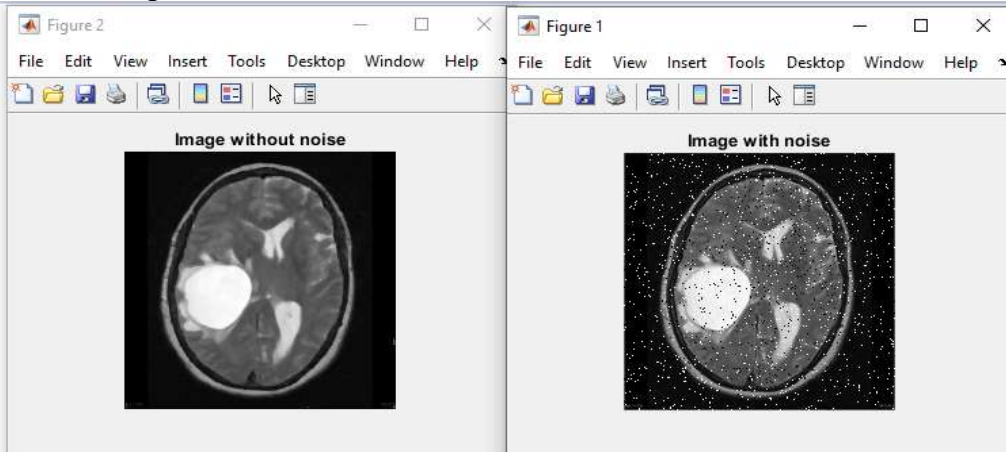


Fig: If image is not selected from the specific folder

2. PRE-PROCESSING:

Pre-processing phase of our project mainly involves those operations that are ordinarily essential before the goal analysis and extraction of the required data and ordinarily geometric corrections of the initial image. These enhancements embrace correcting the information for irregularities and unwanted region noise, removal of non-brain element image and converting the data so that they are correctly reflected in the original image. Major issues related to the pre-processing stage are as follows:-

- a. Noise,
- b. Blur Low Contrast,
- c. Thresholding



This pre-processing stage is used for reducing image noise, highlighting important portions.

a. DE-NOISING:

Here, **Median Filter** is been used as a de-noising technique. This make it possible to recover weak signals from noise and the processed image in this way can be cleaned up without blurring or losing the clarity. And displaying the thresholded image.



b. IMAGE ENHANCEMENT AND FILTERING

The enhancement of the image starts by first converting the gray scale image to black and white image this is done by the use of function **im2bw(gray_image)**. Here the threshold value taken in our project is **0.6**. As Image improvement strategies improve the visual look of pictures from tomography and also the distinction enhancing brain volumes

are linearly associated. Other than this, dilation and erosion has also been performed in this step.



3. FEATURE EXTRACTION:

In this phase the features of the given input image are being extracted. These features include mean, contrast, variance, smoothness, entropy, energy, homogeneity and standard deviation. And on the basis of these features the image is analyzed and the detection of the tumor region is being done.

For Brain edge detection, we have used the **CANNY** method so that internal details of brain are also display, on the other hand, for the tumor boundary, **SOBEL** method is being used that only displays the outline of tumor.



4. SEGMENTATION:

This step involves the conversion of MRI image to binary image with a threshold value of 0.6. Till this step, still some of the white spaces will be left that will be removed in the next analysis steps.

5. IMAGE ANALYSIS:

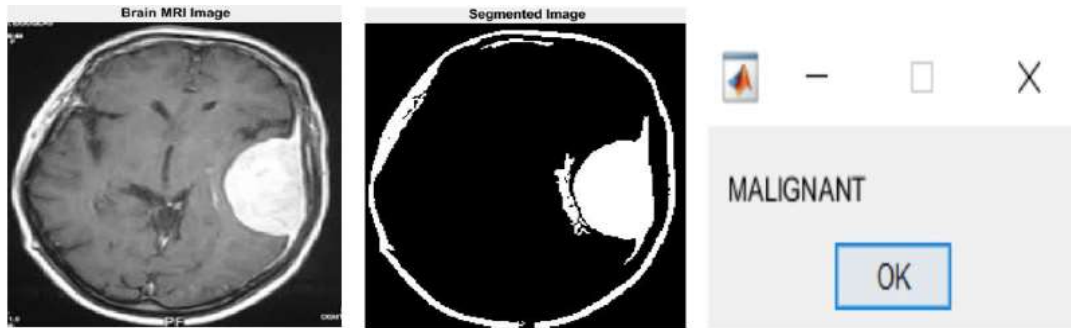
After the tumor is been identified the image analysis is been done to determine the tumor alone and to display the outline of detected tumor. Again these steps are done on the binary image with threshold value of 0.6.



a. CLASSIFICATION:

In this step, we have displayed three major things:

1. **Scan Result** – Either positive or negative
2. **Tumor Result** – Either Benign, Malignant, or No Tumor Found!
3. **Type of Tumor** – Based on the maximum area of the stated tumor



b. PERFORMANCE:

This will contain the execution time, the whole has taken to get completed.

• MATLAB IMPLEMENTATION

o MATLAB CODE

```
% --- Executes on button press in BROWSE.
function BROWSE_Callback(hObject, eventdata, handles)
[FileName,PathName] = uigetfile('*.dcm','Pick an MRI Image: ');
if isequal(FileName,0) || isequal(PathName,0)
    warndlg('Image not Found!!');
else
    I = dicomread([PathName,FileName]);
    info = dicominfo([PathName,FileName])
    I = imresize(I,[200,200]);
end

% --- Executes on button press in PREPROCESSING.
function PREPROCESSING_Callback(hObject, eventdata, handles)
if isfield(handles,'ImgData')
    %if isfield(handles,'imgData')
    I = handles.ImgData;
    %Noise Removal
    Img = rgb2gray(I);
    J = imnoise(Img,'salt & pepper',0.05);
    K = medfilt2(J);
    figure(1)
    imshow(J,[]);
    title('Image with noise')
    figure(2)
    imshow(K,[]);
    title('Image without noise')
    %%Thresholded Image
    image_thresholded = I;
    image_thresholded(I>100) = 0;
    image_thresholded(I<100) = 255;

function DENOISE_Callback(hObject, eventdata, handles)
if isfield(handles,'ImgData')
    %if isfield(handles,'imgData')
    I = handles.ImgData;

    %%Erosion
    se = strel ('line',10,45);
    erodedI = imerode(I,se);

    %%Image Dilation
```

```

        se = strel ('disk',1);
        dilatedI = imdilate(I,se);
%%Image Dilation again
        se = strel ('disk',1);
        dilatedI2 = imdilate(dilatedI,se);
end

% --- Executes on button press in FEATURE.
function FEATURE_Callback(hObject, eventdata, handles)
if isfield(handles, 'ImgData')
    %if isfield(handles, 'imgData')
    I = handles.ImgData;
    Img = rgb2gray(I);

    %%Brain Edge Detection
    b = edge(Img, 'Canny');

    %%Tumor Edge Detection
    bw=im2bw(I,0.6);
    label=bwlabel(bw);
    stats=regionprops(label, 'Solidity', 'Area');
    density=[stats.Solidity];
    area=[stats.Area];
    high_dense_area=density > 0.4;
    max_area=max(area(high_dense_area));
    tumor_label=find(area==max_area);
    tumor=ismember(label,tumor_label);
    se=strel('square',5);
    tumor=imdilate(tumor,se);
    t = edge(tumor, 'Canny');

    stats = graycoprops(Img, 'Contrast Correlation Energy Homogeneity');
    Mean = mean2(I);
    Contrast = stats.Contrast;
    Variance = mean2(var(double(I)));
    sdImage = stdfilt(Img, true(5));
    Smoothness = mean2(sdImage)
    Entropy = entropy(I);
    Energy = stats.Energy;
    Standard_Deviation = std2(I);
    Homogeneity = stats.Homogeneity;
end

% --- Executes on button press in SEGMENTATION.
function SEGMENTATION_Callback(hObject, eventdata, handles)
if isfield(handles, 'ImgData')
    %if isfield(handles, 'imgData')
    I = handles.ImgData;
    img = im2bw(I,0.6);
    img2 = im2bw(I);
    signal1 = img2(:,:);
    axes(handles.axes16)
    imshow(img, []);
end

% --- Executes on button press in ANALYSIS.
function ANALYSIS_Callback(hObject, eventdata, handles)
if isfield(handles, 'ImgData')
    %if isfield(handles, 'imgData')
    I = handles.ImgData;
    bw=im2bw(I,0.6);
    label=bwlabel(bw);
    stats=regionprops(label, 'Solidity', 'Area');
    density=[stats.Solidity];
    area=[stats.Area];
    high_dense_area=density > 0.4;

```

```

max_area=max(area(high_dense_area));
tumor_label=find(area==max_area);
tumor=ismember(label,tumor_label);
se=strel('square',5);
tumor=imdilate(tumor,se);
axes(handles.axes17)
    imshow(tumor,[]);
end

% --- Executes on button press in DETECTION.
function DETECTION_Callback(hObject, eventdata, handles)
if isfield(handles,'ImgData')
    %if isfield(handles,'imgData')
        I = handles.ImgData;
        bw=im2bw(I,0.6);
        label=bwlabel(bw);
        stats=regionprops(label,'Solidity','Area');
        density=[stats.Solidity];
        area=[stats.Area];
        high_dense_area=density > 0.4;
        max_area=max(area(high_dense_area));
        tumor_label=find(area==max_area);
        tumor=ismember(label,tumor_label);
        se=strel('square',5);
        tumor=imdilate(tumor,se);
        [B,L]=bwboundaries(tumor,'noholes');
        axes(handles.axes18)
        imshow(I,[]);
        hold on
        for i=1:length(B)
            plot(B{i}(:,2),B{i}(:,1), 'y' , 'linewidth',1.45);
        end
        title('Detected Tumor');
        hold off;
    end

% --- Executes on button press in CLASSIFICATION.
function CLASSIFICATION_Callback(hObject, eventdata, handles)
if isfield(handles,'ImgData')
    %if isfield(handles,'imgData')
        I = handles.ImgData;
        bw=im2bw(I,0.6);
        label=bwlabel(bw);
        stats=regionprops(label,'Solidity','Area');
        density=[stats.Solidity];
        area=[stats.Area];
        high_dense_area=density > 0.4;
        max_area=max(area(high_dense_area));
        if (max_area < 3000)
            set(handles.result,'string','Benign');
            set(handles.scan,'string','Positive');
            if(max_area<1000 && max_area>500)
                set(handles.type,'string','Glioma');
            elseif(max_area>2500 && max_area<3000)
                set(handles.type,'string','Meningioma');
            elseif(max_area<500 && max_area>0)
                set(handles.type,'string','Craniopharyngioma');
            elseif(max_area>1000 && max_area<2000)
                set(handles.type,'string','Haemangioblastoma');
            else
                set(handles.type,'string','Glioblastoma');
            end
        elseif (max_area > 3000)
            set(handles.result,'string','Malignant');
            set(handles.scan,'string','Positive');
            if(max_area>4000 && max_area<5000)
                set(handles.type,'string','Astrocytoma');
            end
        end
    end
end

```



```

elseif(max_area>5000 && max_area<6000)
    set(handles.type,'string','Craniopharyngioma');
elseif(max_area>6000 && max_area<7000)
    set(handles.type,'string','Oligodendroglioma');
else
    set(handles.type,'string','Ependymoma');
end
else
set(handles.result,'string','No Tumor Found!!');
set(handles.scan,'string','Negative');
    set(handles.type,'string','-');
guidata(hObject,handles);
end
end

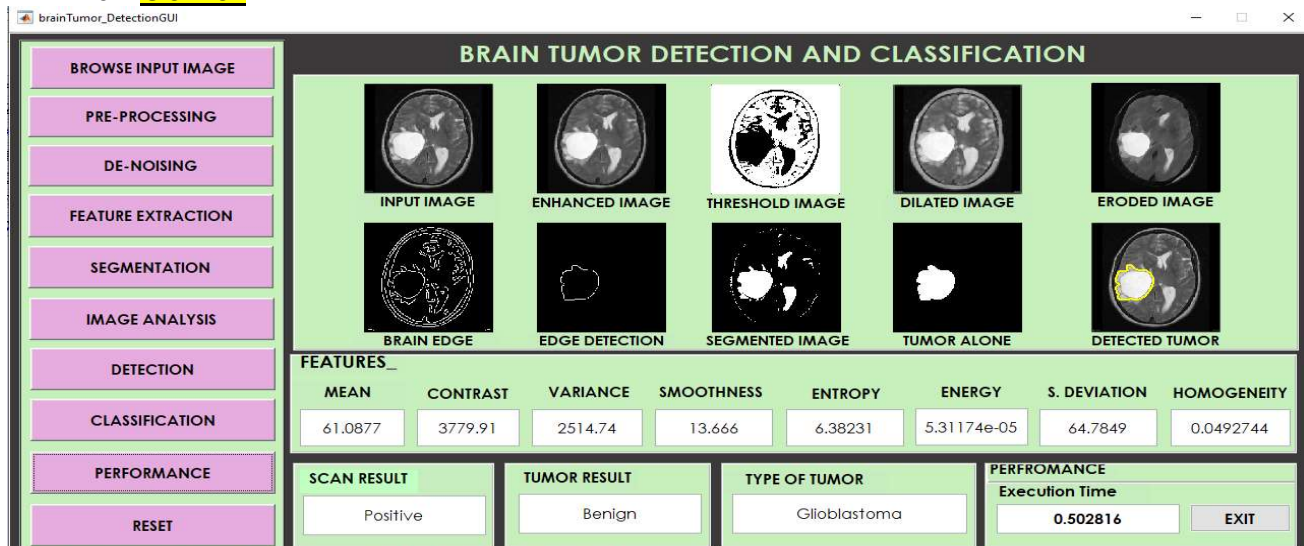
% --- Executes on button press in PERFORMANCE.
function PERFORMANCE_Callback(hObject, eventdata, handles)
tic
pause(0.5);
TIME = toc;
set(handles.TIME,'string', TIME);

% --- Executes on button press in RESET.
function RESET_Callback(hObject, eventdata, handles)
cla(handles.axes10,'reset');
cla(handles.axes11,'reset');
cla(handles.axes12,'reset');
cla(handles.axes13,'reset');
set(handles.MEAN,'string','');
set(handles.CONTRAST,'string','');
set(handles.VARIANCE,'string','');
set(handles.SMOOTH,'string','');
set(handles.ENTROPY,'string','');
set(handles.ENERGY,'string','');
set(handles.SDEV,'string','');
set(handles.HOMO,'string','');
set(handles.scan,'string','');
set(handles.TIME,'string','');
set(handles.result,'string','');
set(handles.type,'string','');

% --- Executes on button press in RESET.
function EXIT_Callback(hObject, eventdata, handles)
close

```

OUTPUT



- **CONCLUSION**

In this project we have automated the diagnosis procedure for the brain tumor detection by the use of image processing. Our project has proved to provide an over all accuracy by upto 97%. All the steps for detecting brain tumor that have been discussed starting from MRI image acquisition ,pre-processing steps to successfully classification of the tumor using the two segmentation techniques have been done. Pre-processing involves operations like wavelet-based methods has been discussed. Quality enhancement and filtering are important because edge sharpening, enhancement, noise removal and undesirable background removal are improved the image quality as well as the detection procedure. Among the different filtering technique, Median filter suppressed the noise without blurring the edges and it is better outlier without reducing sharpness of the images. reduces the noise; enhance the image quality and computationally more efficient than other filtering methodology. After the image quality improvement and noise reduction discussed here, segmentation methodology for a brain tumor from MRI of brain image is been used. Classification based segmentation segment tumor accurately and manufacture sensible results for big information set however undesirable behaviours can occur in case wherever a category is unrepresented in training data. Clustered based segmentation performs is straight forward, quick and manufacture sensible results for non-noise image except for noise pictures it leads to serious inaccuracy within the segmentation. And through these, we have identified and classified the tumor part from rest of the original brain part.
